Hargun Bhatia

Graduate Student, Department of Mathematics, UC San Diego

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Education

University of California San Diego (UCSD), La Jolla, CA
Ph.D. Mathematics — *Sep '21 - Jun '27 (expected)*
Advisor: Claus Sorensen
M.A. Mathematics — *Sep '21 - Mar '23*
GPA: 4.0/4.0

Indian Institute of Technology Kanpur, Uttar Pradesh, India
Bachelor of Science, Mathematics and Scientific Computing — *Jul '17 - May '21*
Mathematics GPA: 9.83/10 Overall GPA: 9.58/10

Research Interests

Number theory and representation theory, particularly smooth mod-p representations of p-adic groups and their connections to the local Langlands program. Specific interests include admissibility and duality in derived categories of smooth representations, differential graded Hecke algebras, and higher mod-p Satake transforms.

Publications and Preprints

- *Higher mod-p Satake transforms: Explicit formulas and images for GL_2* with Yashi Jain, Casey Perdue, Ben Savoie (preprint: arXiv:2610.13203)
- *Cohomology of complex supertori* with Soumya Ganguly, Zhiyuan Jiang, Jeffrey Rabin, Steven Sam (preprint: arXiv:2606.02903)

Talks

- *Representations of p-adic groups*, The San Marcos Informal Mathematics In-Person Colloquium, CSU San Marcos, October 6, 2026.
- *Explicit generalized mod p Satake transforms*, Department Seminar, IIT Kanpur, October 17, 2025.
- *Some problems on the derived category of smooth mod-p representations*, Advancement to PhD Candidacy, UC San Diego, April 23, 2025.
- *Explicit generalized mod p Satake transforms*, Number Theory Seminar, UC San Diego, April 16, 2025.

Teaching and Mentorship

Associate Instructor, UCSD — Summer 2026
- Selected for the Summer Graduate Teaching Scholars (SGTS) program. Independently taught Math 20D (Introduction to Differential Equations), an accelerated five-week course with approximately 25 students. Developed the syllabus and exams, delivered 30 hours of lectures, and coordinated discussion sections with the teaching assistant.

Teaching Assistant, UCSD
- Math 20A (Calculus For Science & Engineering) — Fall 2026
- Math 20D (Introduction to Differential Equations) — Spring 2026
- Math 180A (Introduction to Probability) — Winter 2026
- Math 187A (Introduction to Cryptography) — Fall 2025
- Math 103A (Modern Algebra I) — Summer 2025
- Math 100C (Abstract Algebra III) — Spring 2025
- Math 187A (Introduction to Cryptography) — Winter 2025
- Math 100A (Abstract Algebra I) — Fall 2024
- Math 103B (Modern Algebra II) — Summer 2024
- Math 100C (Abstract Algebra III) — Spring 2024

- Math 103B (Modern Algebra II) — Winter 2024
- Math 103A (Modern Algebra I) — Fall 2023
- Math 103B (Modern Algebra II) — Summer 2023
- Math 187A (Introduction to Cryptography) — Spring 2023, Winter 2023, Fall 2022
- Math 103B (Modern Algebra II) — Summer 2022
- Math 10B (Calculus II) — Spring 2022
- Math 20A (Calculus For Science & Engineering) — Winter 2022, Fall 2021

Instructor, Monsoon Math Camp *Jul '21*
- Co-taught a course on "Proof of Leibniz Formula", which related algebraic and analytic methods in number theory, to a group of highly motivated high school students of India.

Project Mentor, Stamatics, IIT Kanpur *Apr '21 - Jul '21*
- Co-mentored a group of 5 first year undergraduate students on reading the book *Topics in Algebra* by *I.N. Herstein*, gave assignments and conducted weekly problem solving sessions.

Seminar Organization

Cohomology of p-adic Chevalley groups (Co-organized with Casey Perdue) *Fall '25*
Representations of p-adic groups (Co-organized with Casey Perdue) *Fall '24 & Winter '25*
Automorphic Representations (Co-organized with Chris Xu, Online) *Fall '22 & Winter '23*

Conferences and Workshops attended

The Australian Direction, University of Utah, Salt Lake City *Aug '26*
Arithmetic cycles, Modular forms, and L-functions, CRM, Montréal *Aug '25*
Mathematical Congress of the Americas (MCA) 2025, Miami *Jul '25*
Western Algebraic Geometry Symposium (WAGS) 2025, UBC, Vancouver *Apr '25*
Arizona Winter School 2025, University of Arizona, Tucson *Mar '25*
Modular Forms, L-functions, and Eigenvarieties, ENS, Paris *Jun '24*
2024 Lagenhop Lecture & SIU Conference, SIU, Carbondale *May '24*
Arizona Winter School 2024, University of Arizona, Tucson *Mar '24*
Introductory Workshop: Algebraic Cycles, L-Values, and Euler Systems, SLMath, Berkeley *Jan '23*
Southern California Number Theory Day 2022, UC Irvine *Sept '22*
A Pair of Automorphic Workshops (APAW) 2022, University of Oregon, Eugene *Jul '22*
Special Program on p-adic L-functions and Eigenvarieties, University of Notre Dame, Notre Dame *Jul '22*
Conference on Homotopy Theory with applications to Arithmetic and Geometry, The Fields Institute, Toronto *Jun '22*
ICTS workshop on Elliptic Curves and Special Values of L-functions (Online) *Aug '21*
Visiting Students' Research Programme, TIFR, Mumbai *Jun '19*

Honors & Fellowships

James B. Ax Graduate Fellowship, UC San Diego *2021*

General Proficiency Medal, IIT Kanpur *2021*
- For best academic performance in the Department of Mathematics and Statistics

Academic Excellence Award, IIT Kanpur *2018, 2019*
- For outstanding academic performance during each of the academic sessions

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship, Department of Science and Technology, Govt. of India *2017*

National Talent Search Scholarship (NTSE), National Council of Educational Research and Training, Govt. of India *2015*

Languages and Skills

Hindi (Native), English, French (Beginner)

C, C++, Python (NumPy, Pandas, PyTorch), MATLAB, SageMath, LaTeX